

Occlusion Target Detection Based on Improved YOLOv5 Model

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Abstract—To solve the problem that the network is not robust due to the insufficient feature information extraction of occluded objects, an occluded objects detection algorithm combining coordinate attention (CA) mechanism and improved non-maximum suppression (NMS) is proposed in this paper. Based on the YOLOv5 model, the coordinate attention mechanism is first added into the feature extraction layer, which considers both feature information and position information to improve the problem of insufficient feature information extraction due to occlusion. Then, based on Generalized intersection over union (GIoU), the center distance and aspect ratio are considered, Complete-IOU (CIoU) is conducted to replace the original loss function to solve the degradation problem existing in GIoU and the problem that Distance-IOU (DIOU) does not consider aspect ratio. Finally, the traditional NMS only considers Intersection over union (IoU) as an indicator to remove duplicate boxes, which leads to error suppression, kCIoU-NMS is proposed instead of NMS to solve the shortcomings brought by IoU. Tested on a test set with different occlusion rates when the targets are occluded from each other and the background. The accuracy of the model in this paper is above 95% when the occlusion rate is 20% and 40%, and when the occlusion rate is 60%, the detection accuracy is 83.80% and 87.60%, which are 16.00% and 14.40% higher than the original model, respectively. The experimental results show that the method proposed in this paper has good effect and robustness.

Index Terms—Occlusion objects detection, Coordinate attention, CIoU, kCIoU-NMS

I. INTRODUCTION

In real scenes, object detection is still a challenging task because of occlusion. Human eyes can enable humans to judge the attributes of objects by continually inferring through the contours existing in the scene when part of the information of objects is blocked or lost. However, neural networks are not very good at extracting feature information and classifying objects accurately when learning from obscured objects.

The current problem of occlusion object detection suffers from three difficulties: (1) Occlusion interferes with feature extraction, and two targets that are occluded from each other may have very similar characteristics, making it impossible

for the detector to make accurate predictions. The relationship between different channels are found using squeeze and excitation networks (SENet), and the attention-guided network is redesigned to make it more suitable for occlusion detection scenarios [1][2]. However, only the relationship between channels is considered, but the location information is ignored. (2) As there may be serious duplication between prediction boxes when occluded, prediction boxes for different targets may be incorrectly suppressed by the NMS algorithm as predictions for one target, resulting in the occurrence of missed detections. To address the shortcomings of NMS, Bodla et al [3] proposed Soft-NMS, which uses a decaying algorithm to detect all prediction boxes and reduce the probability of boxes being erroneously deleted. Ning et al [4] proposed Weighted NMS, which performs a weighted fusion of prediction box coordinates to obtain high Precision and Recall. Liu et al [5] proposed a density prediction module to improve the accuracy of occluding, but increase the computational effort. A common problem with all the above approaches is that only IoU is considered, which does not provide a comprehensive description of the overlapping relationship of the boxes. (3) Because of missing information due to occlusion, if the loss function is not chosen properly, the network can easily fall into a single direction or local optimum in the process of continuously optimizing the parameters. Rezaatofghi et al [6] proposed GIoU based on IoU and introduced the minimum enclosing box area, solving the problem of IoU but with degradation problems. Zhang and Wang et al [7] proposed DIOU, introducing the distance between two box center points and the minimum enclosing box diagonal length based on GIoU, solving the degradation problem but not considering the aspect ratio.

To address the above problems, this paper proposes an occlusion object detection method based on coordinate attention fusion kCIoU-NMS. Firstly, the coordinate attention mechanism is added to the network structure of YOLOv5 to improve the feature extraction of occluded objects. Then, based on the original loss function GIoU, CIoU was used

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instead of GIoU, considering the distance between the prediction and GT box centers and the aspect ratio. Finally, this paper improves the traditional NMS algorithm and proposes the kCIoU-NMS algorithm, which can not only improve the accuracy of network detection of the occlusion object but also effectively reduce the situation that the traditional NMS has incorrect inference to the occlusion object. The proposed method is validated by applying it to test sets with different occlusion rates and comparing it with existing algorithms. The experimental results show that the proposed method in this paper has good detection results.

II. IMPROVED YOLOV5 ALGORITHM

A. Introduction to The YOLOv5 Model

This paper uses the YOLOv5 network model as the basic model. Its structure is shown in Fig. 1, which includes four parts, namely Input, Backbone, Neck, and Prediction. In this paper, the Input image is scaled to a unified size of 640x640 at the Input end, and Mosaic data enhancement is applied to splice the four images into one image. The Input image is randomly adjusted chroma, saturation, and brightness by the Augment HSV method, and finally, the data is randomly cropped, rotated, panned, and scaled to increase data diversity. The Backbone section is a modified CSP-Darknet53, replacing the Focus block with a 6x6 convolution kernel to ensure that the output is both constant and faster. The SPP block is replaced by the SPPF block, which passes the input through three 5x5 MaxPools in serial, stitching the result of each MaxPool through Concat and feeding it into a 1x1 convolutional kernel to get the output, which has the advantage of keeping the output scale the same compared to the original SPP block but increasing the speed. The Neck part adopts the structure of FPN [8] combined with PAN [9] to fuse the low-level network feature information with the deep network information, which makes the feature information extraction more sufficient and further improves the network detection accuracy. The Prediction part completes the detection of feature maps of different scales according to the number of categories in the dataset.

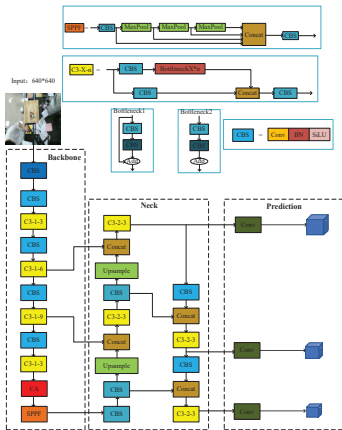


Fig. 1. YOLOv5 structure

B. Coordinate Attention

In this paper, the CA block[10] is added to Backbone and is set at the end of feature extraction to more comprehensively fuse the inter-channel relationships and location information. The CA block has an advantage over the traditional SE block in that it can obtain not only the dependencies between channel relationships but also the location information about the channels, whereas the SE block ignores the location information. the structure of the SE and CA blocks is shown in Fig. 2.

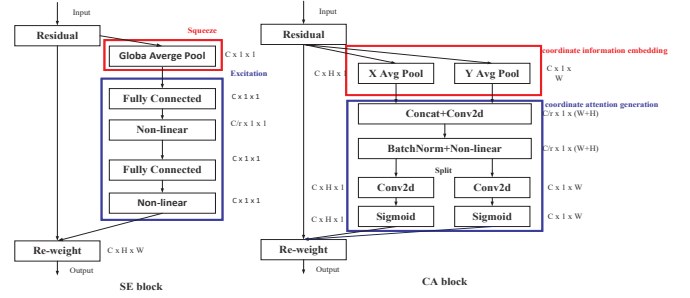


Fig. 2. SE and CA block

The Squeeze operation in the SE block is the use of global average pooling (GAP) after obtaining multiple feature maps, and the area marked in red in Fig. 2 is the Squeeze operation. The equation is as follows (1):

$$z_n = F_{sq}(x_n) = \frac{1}{W \times H} \sum_{i=0}^W \sum_{j=0}^H x_n(i, j) \quad (1)$$

where z_n is the output about the n -th channel after passing the GAP, W and H are the width and height of the feature map, and x_n is the input of the n -th channel. The second step is to fully obtain the importance between channels through the Excitation operation. The equation is as (2):

$$s_n = F_{ex}(z, W) = \sigma(W_2 \delta(W_1 z_n)) \quad (2)$$

where W_1 is a fully connected operation with dimension $\frac{C}{C*r}$ and r is a scaling factor to reduce the number of channels to reduce computation. δ is an activation function. W_2 is a fully connected operation with dimension $\frac{C*r}{C}$, where the number of channels is $C \times 1 \times 1$. σ is a sigmoid activation function.

$$y_n = s_n \cdot x_n \quad (3)$$

Through the output of the sigmoid activation function, s_n is the weight between each channel. The final output y_n after enhancing the feature information through the SE block is obtained by multiplying s_n with the original network input x_n , and the equation is shown in (3).

CA block decomposes channel attention into a 1D feature encoding process that aggregates features along both horizontal and vertical directions. The advantage of this is that not only long-range dependencies can be captured along one spatial direction, but also precise location information in another spatial direction can be captured.

In the first step, coordinate information embedding replaces the original GAP in the SE block with two parallel convolutional kernels, as follows: for input X , pooling operations are performed for each channel along the horizontal and vertical directions using pooling kernels of dimensions $(H, 1)$ and $(1, W)$, respectively, to obtain feature information about the input X in the horizontal and vertical directions. The output of the n -th channel with height h is:

$$z_n^h(h) = \frac{1}{W} \sum_{i=0}^W x_n(h, i) \quad (4)$$

In the same way, the output of the n -th channel of width w is obtained as:

$$z_n^w(w) = \frac{1}{H} \sum_{i=0}^H x_n(i, w) \quad (5)$$

In the first step of embedding the coordinate information, unlike the traditional SE block that performs GAP on the entire feature map, the CA block performs one-directional pooling operations in two directions to obtain features, so that the two The feature information in each direction is fused.

In the second step, the coordinate attention generation uses the two unidirectional feature encoding information generated in the first step to perform a Concat operation on the two encodings, then a convolution operation using a shared 1×1 convolution kernel, and finally an intermediate feature map is obtained by a nonlinear transformation function, as shown in (6):

$$f = \delta(F_1([z^w, z^h])) \quad (6)$$

Where $[\cdot, \cdot]$ represents the Concat operation on the two feature layers along the spatial dimension, F_1 is a 1×1 convolution kernel, and δ is the nonlinear transformation function, which is the ReLu function used in this paper to make nonlinear changes to the output of F_1 .

After that, f was split into two separate tensors $f \in R^{c/r \times H \times 1}$ and $f \in R^{c/r \times 1 \times W}$ along the spatial dimension, and then two 1×1 convolution kernels F_h and F_w are used to expand the number of channels of f^h and f^w to the same as the input X and use the Sigmoid activation function to obtain the weight values g^h and g^w between the range (0, 1), as shown in equations (7) and (8):

$$g^h = \sigma(F_h(f^h)) \quad (7)$$

$$g^w = \sigma(F_w(f^w)) \quad (8)$$

The final output of the CA block can be expressed as equation (9) as follows:

$$y_c(i, j) = x_c(i, j) \times g_c^h(i) \times g_c^w(j) \quad (9)$$

It can be seen from the above derivation that the SE block only considers the relationship between channels to capture the importance of each channel, while the location information is ignored. Location information is very important for the attention mechanism to capture key information, especially for

occlusion targets in a specific working environment, obtaining the coordinate information of the target can often better detect the occlusion target. The CA block encodes the features in two directions, which better preserves the location and feature information of the input feature map compared to the SE block.

C. Loss Function

The loss function is usually used to analyze and evaluate the degree of difference between the true and predicted values of the training model and is positively correlated with the performance of the training model. IoU is the intersection ratio between boxes. The problem with IoU is that when the two boxes do not intersect, it does not reflect the proximity of the two boxes; it does not accurately reflect the degree of overlap between the two boxes, as shown in Fig. 3 below, where the two IoU values are the same, but the degree of overlap is different.

The original YOLOv5 model uses GIoU in the loss function calculation, with the equation shown in equation (11):

$$GIoU = IoU - \frac{|C - A \cup B|}{|C|} \quad (10)$$

where C is the minimum bounding box area of A and B . GIoU introduces the minimum bounding box C based on IoU, which solves the problems existing in IoU, but there are still some problems. When the two boxes are in an inclusive relationship, $C = |A \cup B|$ appears, and GIoU will degenerate into IoU, as shown in Fig. 3 shown.

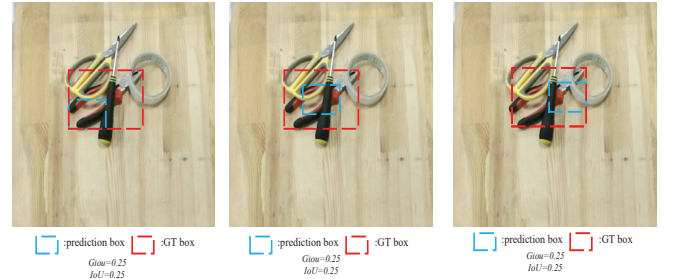


Fig. 3. IoU and GIoU problems

GIoU may have serious errors because obscured objects, there are often many such containing boxes, which when degraded to IoU will not only lead to a decrease in model accuracy, but also a significant decrease in convergence speed, resulting in a long model training time. To improve the degradation problem of GIoU, the center point distance between the prediction box and the GT box can be introduced as a penalty to solve the degradation problem and accelerate the convergence. Therefore, DIoU is introduced, where the equation is as follows:

$$DIoU = IoU - \frac{\rho^2(b, b^{gt})}{c^2} \quad (11)$$

Where b is the coordinates of the center of prediction box A , b^{gt} is the coordinates of the center of GT box B , $\rho^2(b, b^{gt})$ is

the Euclidean distance between the center of A and the center of B , and c is the diagonal distance of the minimum external box. DIoU is calculated as shown in Fig. 4.

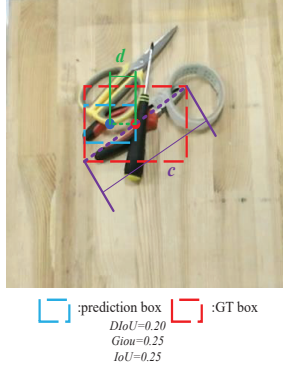


Fig. 4. DIoU problems

DIoU solves the degradation problem by considering the center point coordinates and diagonal distances based on GIoU, but does not take into account the relationship between the aspect ratio of the prediction box and the GT box. Therefore, CIoU is introduced, and CIoU adds the influence factor on top of DIoU, as shown in equations (12)-(13):

$$CIoU = DIoU - \alpha\gamma \quad (12)$$

$$\begin{cases} \alpha = \frac{\gamma}{1 - IoU + \gamma} \\ \gamma = \frac{4}{\pi} (\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h}) \end{cases} \quad (13)$$

Where α is the balance factor, γ is a measure of aspect ratio, w and h are the width and height of the prediction box, w^{gt} and h^{gt} are the width and height of the GT box. By adding the influence factor $\alpha\gamma$ to DIoU, the three elements of intersection and merge ratio, center distance, and aspect ratio are taken into account, which is more closely matched to the GT box compared to DIoU. Therefore, this paper adopts CIoU as the loss function and calculates the loss value as in equation (14):

$$L_{CIoU} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha\gamma \quad (14)$$

D. kCIoU-NMS

Traditional object detection for the removal of duplicate boxes uses NMS, using only IoU as an indicator for redundant boxes, with the following NMS equation:

$$s_i = \begin{cases} s_i, IoU(B_1, B_i) < \varepsilon \\ 0, IoU(B_1, B_i) \geq \varepsilon \end{cases} \quad (15)$$

where B_1 represents the detection box with the highest classification score, B_i represents the i -th detection box compare to B_1 , s_i represents the classification score of B_i , and ε represents the threshold value. When $IoU(B_1, B_i) \geq \varepsilon$, delete B_i , and so on until the IoU between all remaining boxes and B_1 is not greater than ε . Then select the second-highest scoring detection box B_2 and repeat the above process until the IoU between all boxes is less than ε .

Traditional NMS can lead to missed detections when targets obscure each other or detect target crowding, and the missed

detections increase when the IoU is larger the closer the targets are to each other. Therefore, according to the research background, this paper improves the traditional NMS. In combination with equation (12), kCIoU is added to the NMS. kCIoU has the advantage of considering not only the IoU between two detected boxes but also the center distance between the boxes, the aspect ratio, and the classification of the boxes to create the following equation.

$$P_{CIoU} = \frac{\rho^2(b, b^{gt})}{c^2} + k\alpha\gamma \quad (16)$$

Where ρ represents the distance between the center points of A and B , and $\alpha\gamma$ is the equation (13). When A and B belong to the same class, $k = 1$. When A and B do not belong to the same class, $k = 0$. When the categories are different, the aspect ratio of the two detection boxes is too different, which eventually leads to the calculation being very unstable, and there will be negative values or values far greater than 1. Therefore, the score of the classification box based on kCIoU-NMS is obtained as follows

$$s_i = \begin{cases} s_i, P_{kCIoU}(B_1, B_i) < \varepsilon \\ 0, P_{kCIoU}(B_1, B_i) \geq \varepsilon \end{cases} \quad (17)$$

The inclusion of kCIoU makes the selection of thresholds less demanding and takes into account the distance between the center of the redundant box and the detected box with the highest score as well as the aspect ratio, which helps to alleviate the phenomenon of false suppression due to the target occlusion, as shown in Fig. 5 below:

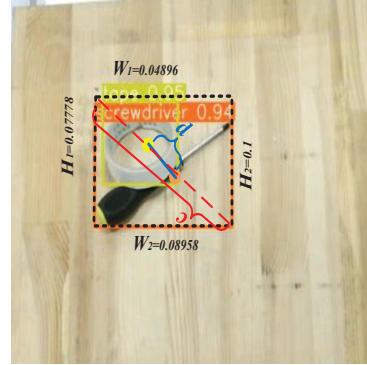


Fig. 5. kCIoU calculation method

The dimensions marked in Fig. 5 are normalized values for a 1920*1080 size image input, with $IoU=0.425$ calculated by traditional NMS, and $s=0.385$ calculated by kCIoU-NMS. When ε is chosen to be 0.42, the traditional NMS will delete the Screwdriver detection box, resulting in poorer detection. kCIoU-NMS can mitigate the phenomenon of false detection due to occlusion by using the center point distance and aspect ratio, which can make the detection effect more realistic.

III. EXPERIMENT AND RESULTS

A. Experimental Environment

The computer system is Windows 10, the CPU is an Inter Core i5 Hexa-core, and the GPU is an NVIDIA GeForce RTX

3080 (12GB). The deep learning framework is Pytorch, and the images are captured using Kinect 2.0.

B. DataSet

A total of 3250 images were collected as training data, of which 2750 were the training set and 500 were the validation set. The data set contains a total of 5 detection targets, namely plier, scissors, screwdriver, solidgum, and tape, where each target contains 1147, 1061, 1102, 1068, and 1132 labels in the training set. The training image with batch size=16 was obtained by Mosaic data enhancement, Augment HSV to adjust the chromaticity, saturation, lightness, and darkness, and after rotating, panning, and scaling operations as shown in Fig. 6 below



Fig. 6. Image after data enhancement

To verify the accuracy of the model, the test set where the target objects occlude each other and the background objects occlude the target under different occlusion rates are collected as verification, in which the target objects occlude each other by 20%-40%, 40%-60%, 60%-80%. The test set below is 600 pieces each. The background objects occlude the target also at 20%-40%, 40%-60%, and 60%-80%, and collect 500 images each as the test set.

C. Evaluation Indicators

Miss detection rate, false detection rate, and accuracy rate are used as evaluation indicators. The false detection rate indicates when the number of detected boxes in the test image is greater than the number of GT boxes, the missed detection rate indicates when the number of detected boxes in the test image is less than the number of GT boxes, and the accuracy rate indicates the probability that the entire test set is identified correctly. The equation for the false detection rate and the missed detection rate is shown below.

$$\begin{cases} F = \frac{\sum_i^N T_{Fi}}{\sum_i^N T_{Mi}} \\ M = \frac{\sum_i^N T_{Mi}}{\sum_i^N T_{Mi}} \\ Ac = 1 - (F + M) \end{cases} \quad (18)$$

$T_{Fi} = 1$ represents a false detection on the i -th image, $T_{Mi} = 1$ represents a miss detection on the i -th image, N is the total number of images tested, F , M and Ac stand for false detection rate, missed detection rate and accuracy rate respectively.

D. Experimental Result

After the training set and the validation set are trained by different models, the epoch is set 150 times, and the results of each model in the test set are shown in Table 1. The comparison of the loss values of the training set and the validation set in the training process of the original model and the model in this paper is shown in Fig. 7, and the comparison of mAP@0.5 and mAP@0.5:0.95 results are shown in Fig. 8:

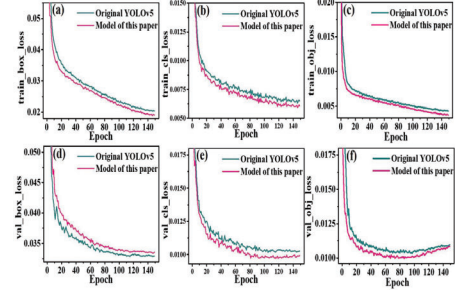


Fig. 7. Comparison of training and validation losses

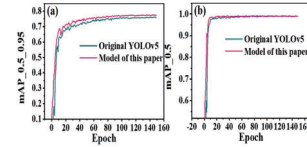


Fig. 8. Comparison of mAP@0.5:0.95 and mAP@0.5

From Fig. 7(a)-(c) and (d)-(f), it can be seen that the model in this paper, after adopting CIOU as the loss function for box loss, has lower box loss, cls loss, and obj loss than the original YOLOv5 training as well as a validation process, indicating that the prediction results of the model in this paper are closer to the real target during the training process. Fig. 8 is obtained from the results of mAP@0.5:0.95. The original model has a maximum mAP@0.5:0.95=0.7610, and the model in this paper has a maximum mAP@0.5:0.95=0.7765. At the same time, the average mAP of the model in this paper is higher than the original model in different epochs, indicating that this paper The model has better detection accuracy.

TABLE I
RESULTS FOR DIFFERENT OCCLUSION RATES BETWEEN TARGETS

Osc rate	module	Ac of TO	M of TO	F of TO
20%	original model	92.33%	0.00%	7.67%
	yolov5+SE	88.83%	0.33%	10.83%
	yolov5+CA	93.67%	0.17%	6.17%
	yolov5+CA+DIoU	96.33%	0.00%	3.83%
	yolov5+CA+CIoU	98.17%	0.00%	1.83%
	This paper	99.00%	0.00%	1.00%
40%	original model	89.83%	0.67%	9.50%
	yolov5+SE	85.67%	1.33%	13.00%
	yolov5+CA	90.00%	1.50%	8.50%
	yolov5+CA+DIoU	90.67%	0.67%	8.67%
	yolov5+CA+CIoU	94.50%	1.33%	4.17%
	This paper	95.00%	0.83%	4.17%
60%	original model	67.80%	17.60%	14.60%
	yolov5+SE	66.20%	14.60%	19.20%
	yolov5+CA	72.40%	17.20%	10.60%
	yolov5+CA+DIoU	74.80%	15.80%	9.40%
	yolov5+CA+CIoU	78.60%	18.60%	2.80%
	This paper	83.80%	13.20%	3.00%

TABLE II
RESULTS FOR DIFFERENT OCCLUSION RATES BETWEEN BACKGROUND

Osc rate	module	Ac of BG	M of BG	F of BG
20%	original model	97.60%	2.00%	0.40%
	yolov5+SE	98.00%	1.60%	0.40%
	yolov5+CA	98.40%	1.40%	0.20%
	yolov5+CA+DIoU	98.60%	1.40%	0.00%
	yolov5+CA+CIoU	99.20%	0.60%	0.20%
	This paper	99.60%	0.20%	0.20%
40%	original model	86.60%	7.20%	6.20%
	yolov5+SE	91.20%	3.60%	5.20%
	yolov5+CA	91.00%	4.20%	4.80%
	yolov5+CA+DIoU	92.00%	4.20%	3.80%
	yolov5+CA+CIoU	93.90%	2.80%	3.40%
	This paper	97.00%	1.40%	1.60%
60%	original model	73.20%	14.00%	12.80%
	yolov5+SE	78.40%	9.60%	12.00%
	yolov5+CA	78.80%	9.00%	12.20%
	yolov5+CA+DIoU	82.80%	8.20%	9.00%
	yolov5+CA+CIoU	86.80%	6.80%	6.40%
	This paper	87.60%	5.40%	7.00%



Fig. 9. Contrast of Target Occlusion

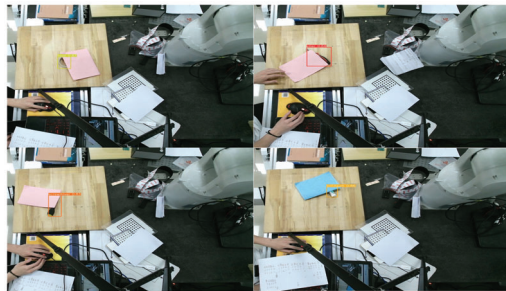


Fig. 10. Contrast of Background Occlusion

Table 1 represents the test results under different occlusion rates when the target objects are occluded from each other, where TO denotes Target Obscuring. when the occlusion rate reaches 60%, the model in this paper achieves 83.80% accurate recognition rate, which is 16.00% better than the original model, with 11.60% less false detection rate and 4.40% less missed detection rate. BO denotes Background Obscuring, and the results in Table 2 demonstrate that this model is still effective in non-target obscuring, with 14.40% better accuracy, 8.60% less missed detection, and 5.80% less false detection compared to the original model at 60% obscuring. Fig. 9 shows a comparison of the detection results of targets occluding each other, the original model on the left misses the Tape and the

improved model on the right identifies the Tape at over 60% occlusion rate. Fig. 10 shows the detection results for different objects at 60% background occlusion rate. As can be seen from Fig. 10, the recognition confidence level still exceeds 80% at severe occlusion, indicating the robustness of the model.

IV. CONCLUSIONS

To improve the detection accuracy of occluded targets and reduce the rate of missed and false detections, this paper proposes an occlusion target detection model based on the improved YOLOv5. The CA block is used to extract the relationship between feature map channels and coordinate information to enhance the network feature information extraction. CIoU is also used as the loss function, taking into account not only the overlap between boxes but also the center point distance and the aspect ratio. For the problem that the traditional NMS will erroneously remove prediction boxes that are heavily overlapped due to occlusion, kCIoU-NMS is proposed to improve the way of removing redundant boxes by considering the center distance between different targets, the diagonal distance of the minimum enclosing box, and the aspect ratio relationship between different kinds of targets.

Also, under the test set with severe occlusion, the comparison of the effect is more obvious, the accuracy of the model in this paper is above 95% at a 20% occlusion rate and 40% occlusion rate. When a 60% occlusion rate, the detection accuracy is 83.80% and 87.60%, the present model is more robust compared to other models.

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